Cambridge (CIE) A Level Maths: Pure 1

Your notes

Applications of Differentiation

Contents

- Gradients, Tangents & Normals
- Increasing & Decreasing Functions
- Second Order Derivatives
- Stationary Points & Turning Points
- Sketching Gradient Functions
- Modelling with Differentiation (Optimisation)
- Connected Rates of Change

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Your notes

Gradients, tangents & normals

How do I find the gradient at a point on a curve?

- To find the gradient of a curve **y= f(x)** at any point on the curve, substitute the **x-coordinate** of the point into the derivative **f'(x)**

e.g. FIND THE GRADIENT OF THE CURVE $y = x - \frac{1}{x}$ AT THE POINT $(2, \frac{3}{2})$

$y = x - x^{-1}$

$\frac{dy}{dx} = 1-(-1)x^{-1-1} = 1+x^{-2} = 1 + \frac{1}{x^2}$

FIND DERIVATIVE

WHEN $x = 2$,

$\frac{dy}{dx} = 1 + \frac{1}{2^2} = \frac{5}{4}$

PUT x-COORDINATE INTO DERIVATIVE

THE GRADIENT AT $(2, \frac{3}{2})$ IS $\frac{5}{4}$

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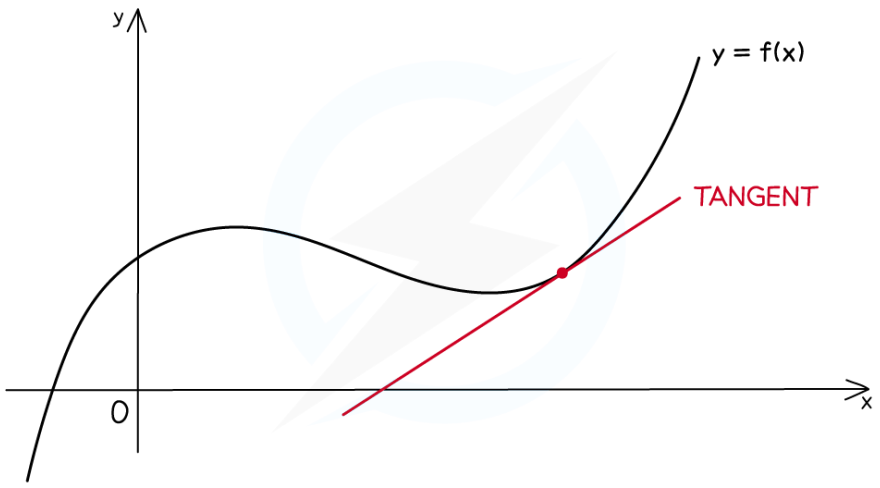
How do I find the equation of a tangent to a curve?

- At any point on a curve, the **tangent** is the line that goes through the point and has the same gradient as the curve at that point



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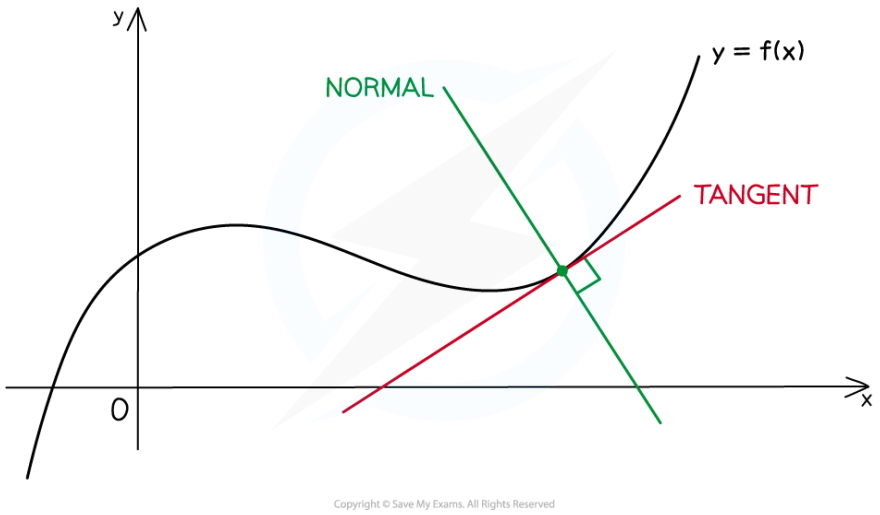
- For the curve y = f(x), you can find the equation of the **tangent** at the point (a, f(a)) using
$$y - f(a) = f'(a)(x - a)$$

Your notes

How do I find the equation of a normal to a curve?

- At any point on a curve, the **normal** is the line that goes through the point and is **perpendicular** to the tangent at that point



- For the curve y = f(x), you can find the equation of the **normal** at the point (a, f(a)) using
$$y - f(a) = -\frac{1}{f'(a)}(x - a)$$

Examiner Tips and Tricks

- The formulae above are not in the exam formulae booklet, but if you understand what tangents and normals are, then the formulae follow from the equation of a straight line combined with parallel and perpendicular gradients (see Worked Example below).

Worked Example

Your notes

The curve C has the equation $y = 7 - 12\sqrt{x}$.

a) Find the equation of the tangent to C at the point where $x = 9$.

b) Find the equation of the normal to C at the same point.

a) WHEN $x = 9$, $y = 7 - 12\sqrt{9} = -29$ ← FIND y-COORDINATE

$y = 7 - 12x^{\frac{1}{2}}$ ← $\sqrt{x} = x^{\frac{1}{2}}$

$\frac{dy}{dx} = -12\left(\frac{1}{2}\right)x^{\frac{1}{2}-1} = -6x^{-\frac{1}{2}} = -\frac{6}{\sqrt{x}}$ ← FIND DERIVATIVE

WHEN $x = 9$,

$\frac{dy}{dx} = -\frac{6}{\sqrt{9}} = -2$ ← USE DERIVATIVE TO FIND GRADIENT OF TANGENT

SO THE TANGENT AT POINT $(9, -29)$ HAS EQUATION

$y - (-29) = -2(x - 9)$ ← $y - y_1 = m(x - x_1)$

$y = -2x - 11$

b) THE NORMAL IS PERPENDICULAR TO THE TANGENT, SO THE GRADIENT OF THE NORMAL IS $-\frac{1}{(-2)} = \frac{1}{2}$

$m_2 = -\frac{1}{m_1}$ FOR PERPENDICULAR LINES

AT $(9, -29)$ THE NORMAL HAS EQUATION

$y - (-29) = \frac{1}{2}(x - 9)$

$y = \frac{1}{2}x - \frac{67}{2}$

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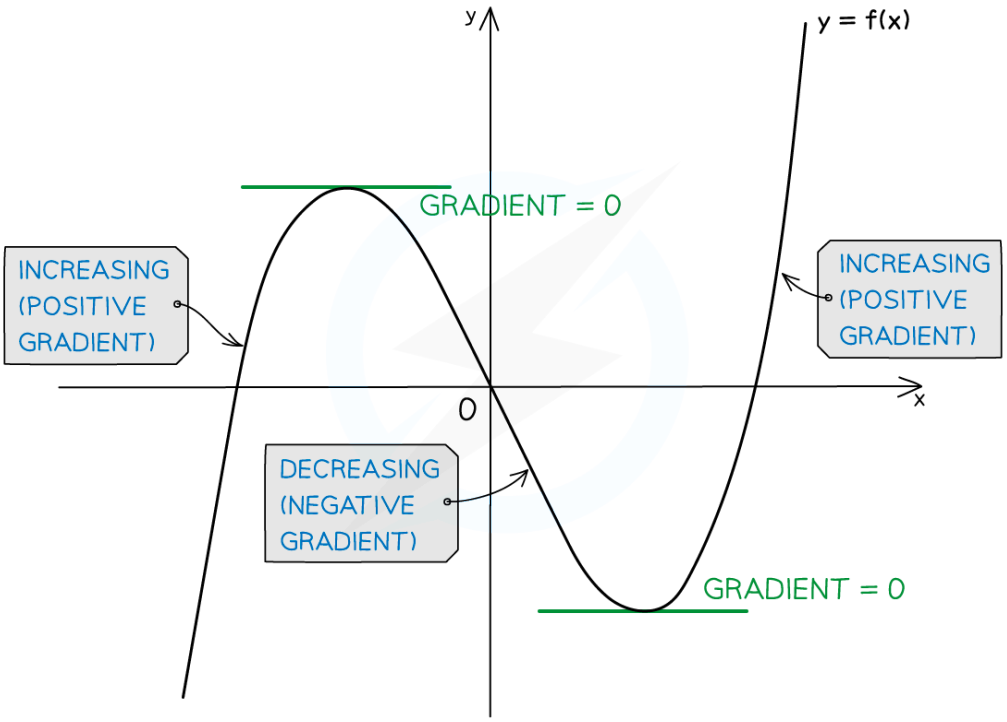
Your notes

Increasing & decreasing functions

What are increasing and decreasing functions?

- A function **f(*x*)** is **increasing** on an interval **[*a*, *b*]** if **f'(*x*) ≥ 0** for all values of ***x*** such that ***a*< *x* < *b***.
  - If **f'(*x*) > 0** for all ***x*** values in the interval then the function is said to be **strictly increasing**
  - In most cases, on an increasing interval the graph of a function goes **up** as ***x*** increases
- A function **f(*x*)** is **decreasing** on an interval **[*a*, *b*]** if **f'(*x*) ≤ 0** for all values of ***x*** such that ***a*** **< *x* < *b***
  - If **f'(*x*) < 0** for all ***x*** values in the interval then the function is said to be **strictly decreasing**
  - In most cases, on a decreasing interval the graph of a function goes **down** as ***x*** increases



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- To identify the intervals on which a function is increasing or decreasing you need to:

1. Find the derivative **f'(*x*)**
2. Solve the inequalities **f'(*x*) ≥ 0** (for increasing intervals) and/or **f'(*x*) ≤ 0** (for decreasing intervals)

Your notes

Examiner Tips and Tricks

- On an exam, if you need to show a function is increasing or decreasing you can use either strict (<, >) or non-strict (≤, ≥) inequalities

Worked Example

Your notes

$f(x) = x^3 + x^2 - x + 2$

a) Find the values of x for which $f(x)$ is an increasing function.

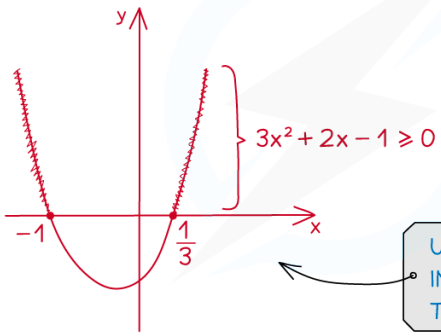
b) Find the values of x for which $f(x)$ is a decreasing function.

a) $f'(x) = 3x^2 + 2x - 1$ ← FIND THE DERIVATIVE

$3x^2 + 2x - 1 = 0$

$(3x-1)(x+1) = 0$

$x = \frac{1}{3}$ OR $x = -1$ ← SOLVE THE QUADRATIC



USE SKETCH TO IDENTIFY INTERVALS SATISFYING THE INEQUALITY

f(x) IS INCREASING WHERE $f'(x) \geqslant 0$

SO f(x) IS INCREASING ON THE INTERVALS $(-\infty, -1]$ AND $[\frac{1}{3}, +\infty)$

b) f(x) IS DECREASING WHERE $f'(x) \leqslant 0$

SO f(x) IS DECREASING ON THE INTERVAL $[-1, \frac{1}{3}]$

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Second order derivatives

What is the second order derivative of a function?

- If you differentiate the derivative of a function (ie differentiate the function a second time) you get the **second order derivative** of the function
- For a function y = f(x), there are two forms of notation for the second derivative (or second order derivative)

$$f''(x) \text{ or } \frac{d^2y}{dx^2}$$

- Note the positions of the power of 2's in the second version

e.g. $f(x) = x^3$

$f'(x) = 3x^2$

$f''(x) = 6x$

$y = 7x^2 - 4x + 19$

$\frac{dy}{dx} = 14x - 4$

$\frac{d^2y}{dx^2} = 14$

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- The second order derivative can also be referred to simply as the **second derivative**
  - Similarly, the 'regular' derivative can also be referred to as either the **first order derivative** or the **first derivative**
- The second order derivative gives the rate of change of the gradient function (ie of the first derivative) – this will be important for identifying the nature of **stationary points**

Examiner Tips and Tricks

- When finding second derivatives be especially careful with functions that have negative or fractional powers of ***x*** (see Worked Example below).

- Mistakes made with fractions or negative signs can build up as you calculate the derivative more than once.

Your notes

Worked Example

Given that $f(x) = 4 - \sqrt{x} + \dfrac{3}{\sqrt{x}}$

a) Find $f'(x)$ and $f''(x)$.

b) Evaluate $f''(3)$. Give your answer in the form $a\sqrt{b}$, where b is an integer and a is a rational number.

a) $f(x) = 4 - x^{\frac{1}{2}} + 3x^{-\frac{1}{2}}$

REWRITE AS POWERS OF x

$f'(x) = -(\frac{1}{2})x^{\frac{1}{2}-1} + 3(-\frac{1}{2})x^{-\frac{1}{2}-1}$

$f'(x) = -\frac{1}{2}x^{-\frac{1}{2}} - \frac{3}{2}x^{-\frac{3}{2}}$

DIFFERENTIATE ONCE TO FIND $f'(x)$

$f''(x) = -\frac{1}{2}(-\frac{1}{2})x^{-\frac{1}{2}-1} - \frac{3}{2}(-\frac{3}{2})x^{-\frac{3}{2}-1}$

$f''(x) = \frac{1}{4}x^{-\frac{3}{2}} + \frac{9}{4}x^{-\frac{5}{2}}$

DIFFERENTIATE A SECOND TIME TO FIND $f''(x)$

b) $f''(x) = \dfrac{1}{4x\sqrt{x}} + \dfrac{9}{4x^2\sqrt{x}}$

$x^{\frac{3}{2}} = x\sqrt{x}$ $\quad x^{\frac{5}{2}} = x^2\sqrt{x}$

$f''(3) = \dfrac{1}{12\sqrt{3}} + \dfrac{9}{36\sqrt{3}}$

$= \dfrac{12}{36\sqrt{3}} = \dfrac{1}{3\sqrt{3}} \times \dfrac{\sqrt{3}}{\sqrt{3}} = \dfrac{\sqrt{3}}{9}$

$f''(3) = \frac{1}{9}\sqrt{3}$

RATIONALISE DENOMINATOR

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Your notes

Stationary points & turning points

What are stationary points?

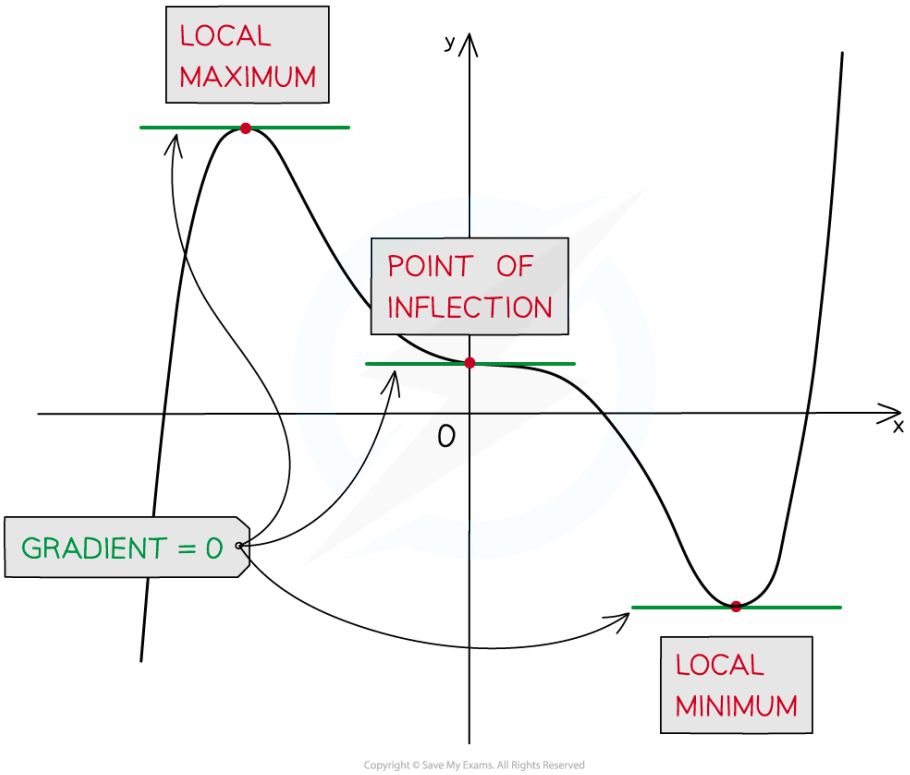
- A stationary point is any point on a curve where the gradient is zero
- To find stationary points of a function **f(x)**

Step 1: Find the first derivative **f'(x)**

Step 2: Solve **f'(x) = 0** to find the *x*-coordinates of the stationary points

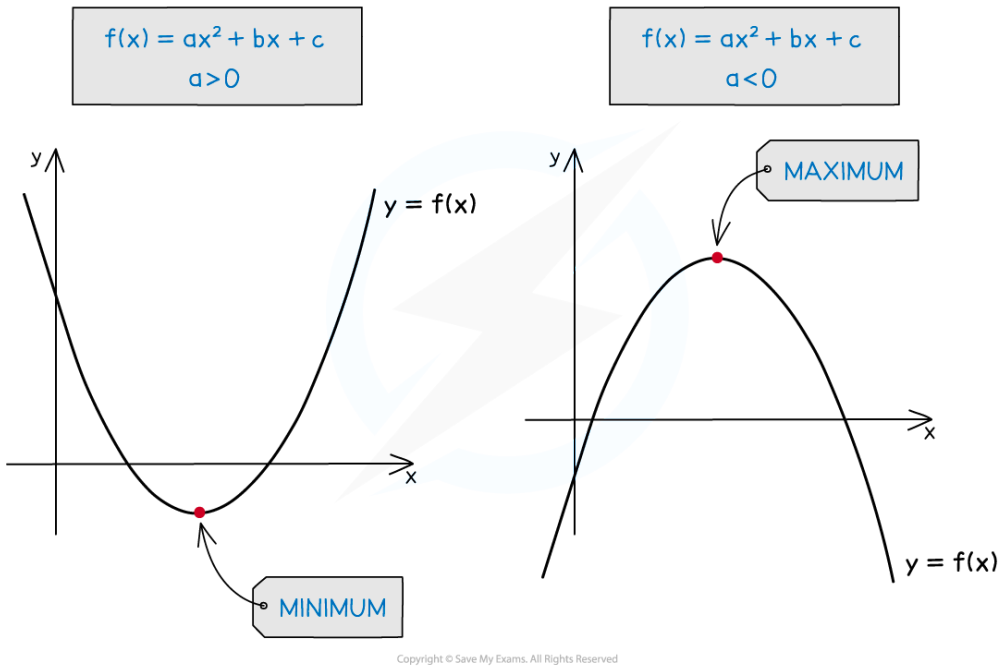
Step 3: Substitute those *x*-coordinates into **f(x)** to find the corresponding *y*-coordinates

- A stationary point may be either a **local minimum**, a **local maximum**, or a **point of inflection**
- **Turning points** are maximum points or minimum points only
  - The graph turns from going up to down (or down to up)



How do I determine the type of stationary point on a quadratic?

- The graph of a quadratic function (ie a parabola) only has a single stationary point
- For an 'up' parabola this is the **minimum**; for a 'down' parabola it is the **maximum** (no need to talk about 'local' here)



- The ***y*** value of the stationary point is thus the minimum or maximum value of the quadratic function
- Quadratics can only have minimum and maximum points
  - so quadratics have **turning points** (also called a **vertex**)

How do I determine the type of stationary point on a curve?

- For a function **f(*x*)** there are two ways to determine the nature of its stationary points
- **Method A:** Compare the signs of the **first derivative** (positive or negative) a little bit **to either side of the stationary point**
  - (After completing **Steps 1 – 3 above** to find the stationary points)

Step 4: For each stationary point find the values of the first derivative a little bit 'to the left' (ie slightly smaller *x* value) and a little bit 'to the right' (slightly larger *x* value) of the stationary point

Your notes

Your notes

- CHOOSING x VALUES ONE LESS AND ONE MORE THAN THE x-COORDINATE OF THE STATIONARY POINT WILL USUALLY WORK HERE (SEE THE EXAMPLE AT THE END)
- DO CHOOSE A CONVENIENT POINT WHERE POSSIBLE ($x = 0$ AND $x = 1$ ARE BOTH VERY EASY TO PUT INTO FORMULAS FOR EXAMPLE)
- DO NOT HOWEVER CHOOSE AN x VALUE TO THE RIGHT OR LEFT THAT JUMPS PAST THE x-COORDINATE OF ANOTHER STATIONARY POINT!

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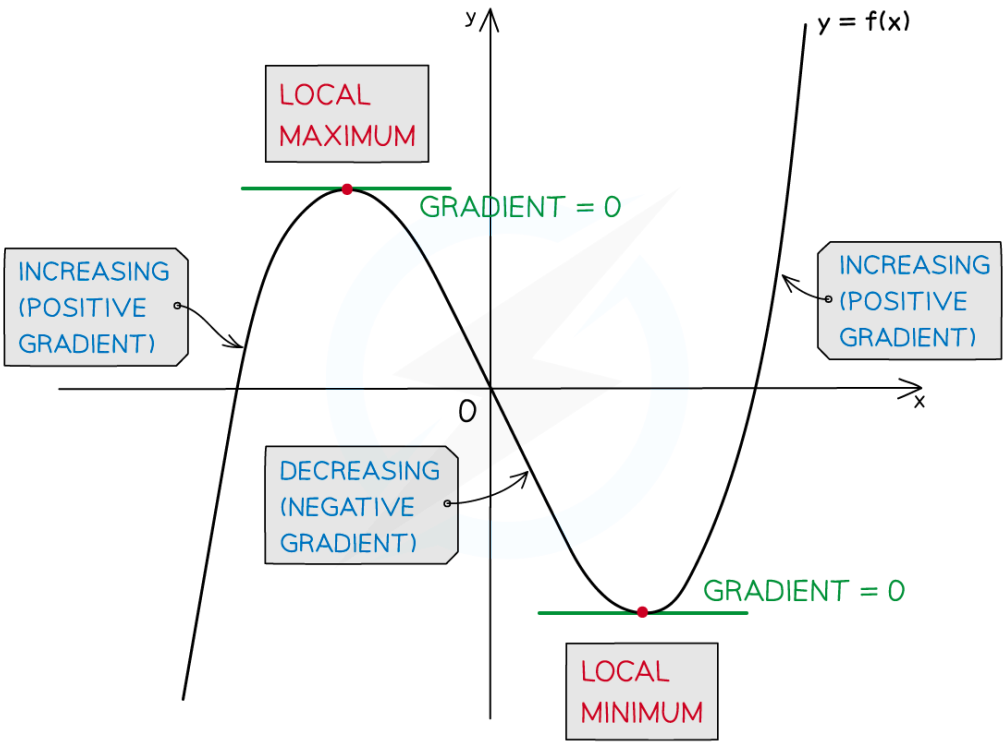
Step 5: Compare the **signs** (positive or negative) of the derivatives on the left and right of the stationary point

- If the derivatives are **negative** on the **left** and **positive** on the **right**, the point is a **local minimum**
- If the derivatives are **positive** on the **left** and **negative** on the **right**, the point is a **local maximum**
- If the signs of the derivatives are the **same** on both sides (both positive or both negative) then the point is a **point of inflection**



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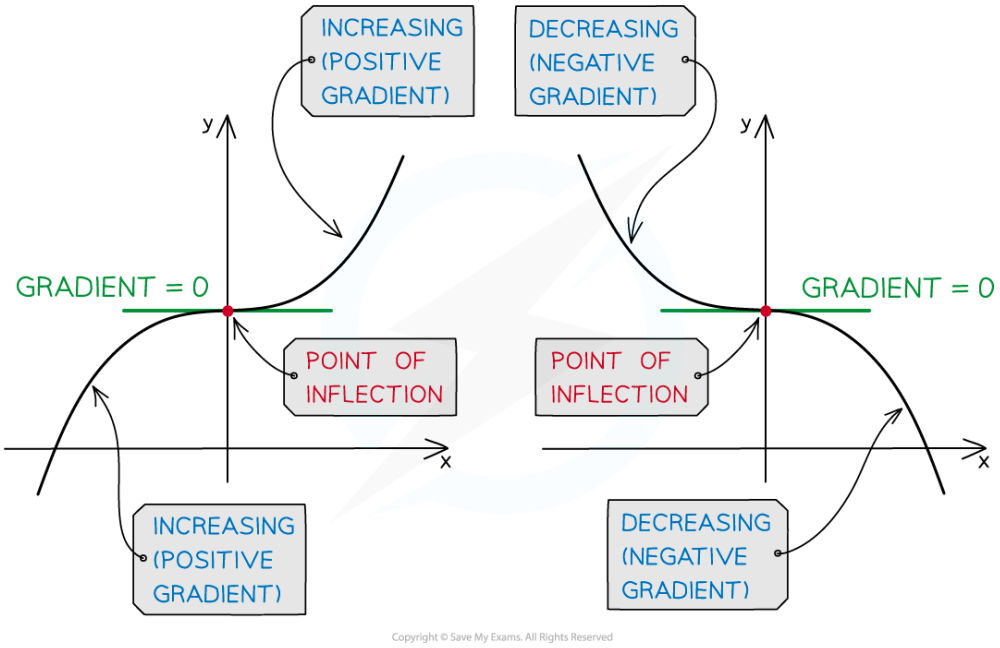
INCREASING (POSITIVE GRADIENT)
DECREASING (NEGATIVE GRADIENT)
y
y
GRADIENT = 0
GRADIENT = 0
POINT OF INFLECTION
POINT OF INFLECTION
x
x
INCREASING (POSITIVE GRADIENT)
DECREASING (NEGATIVE GRADIENT)
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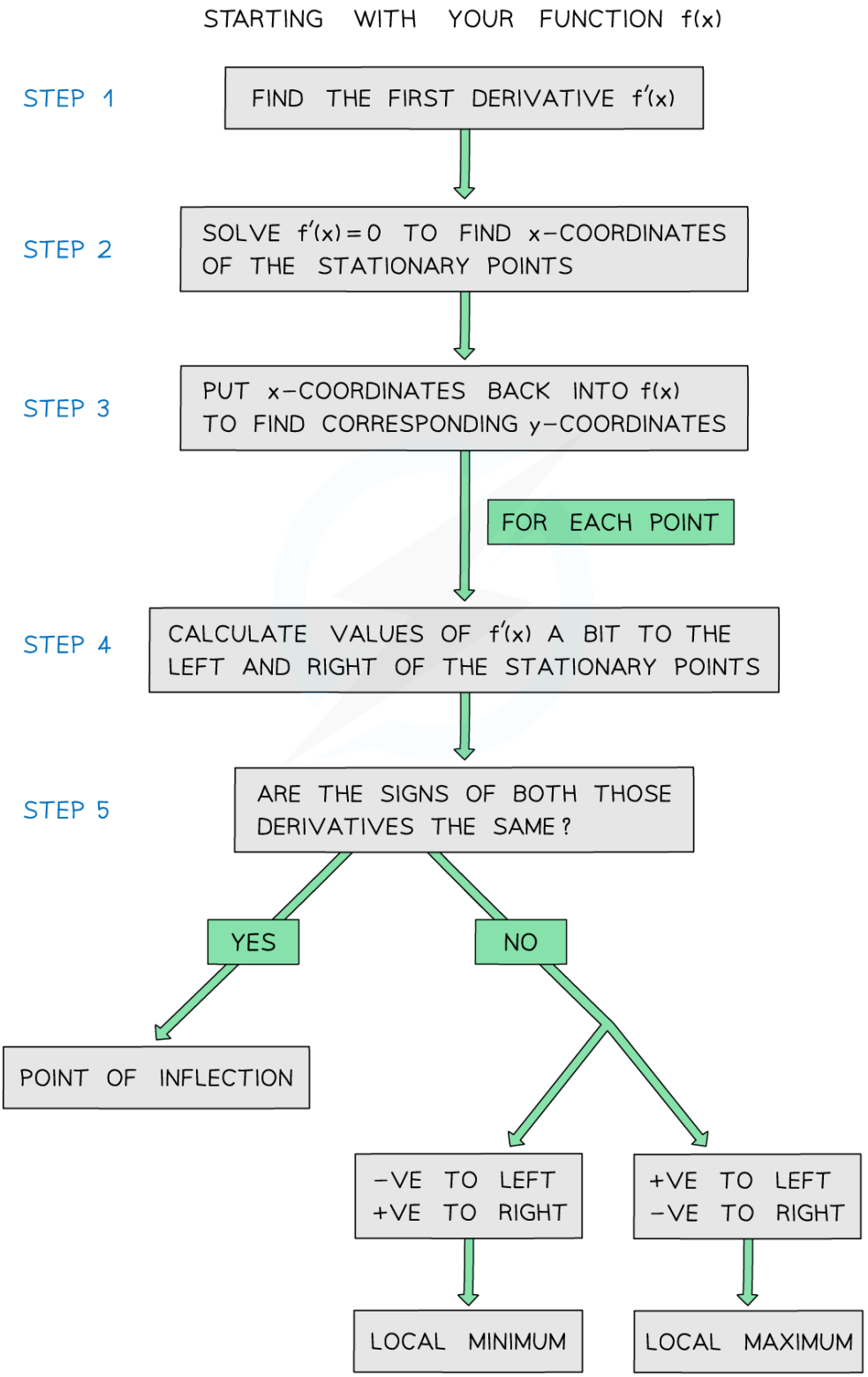
Your notes

METHOD A: DETERMINING NATURE OF STATIONARY POINTS USING FIRST DERIVATIVE



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Method B: Look at the sign of the **second derivative** (positive or negative) **at the stationary point**

(After completing **Steps 1 – 3 above** to find the stationary points)

Step 4: Find the second derivative **f''(*x*)**

Step 5: For each stationary point find the value of **f''(*x*)** at the stationary point (ie substitute the *x*-coordinate of the stationary point into **f''(*x*)**)

- If **f''(*x*)** is **positive** then the point is a **local minimum**
- If **f''(*x*)** is **negative** then the point is a **local maximum**
- If **f''(*x*)** is **zero** then the point could be a **local minimum**, a **local maximum** OR a **point of inflection** (use Method A to determine which)

Your notes

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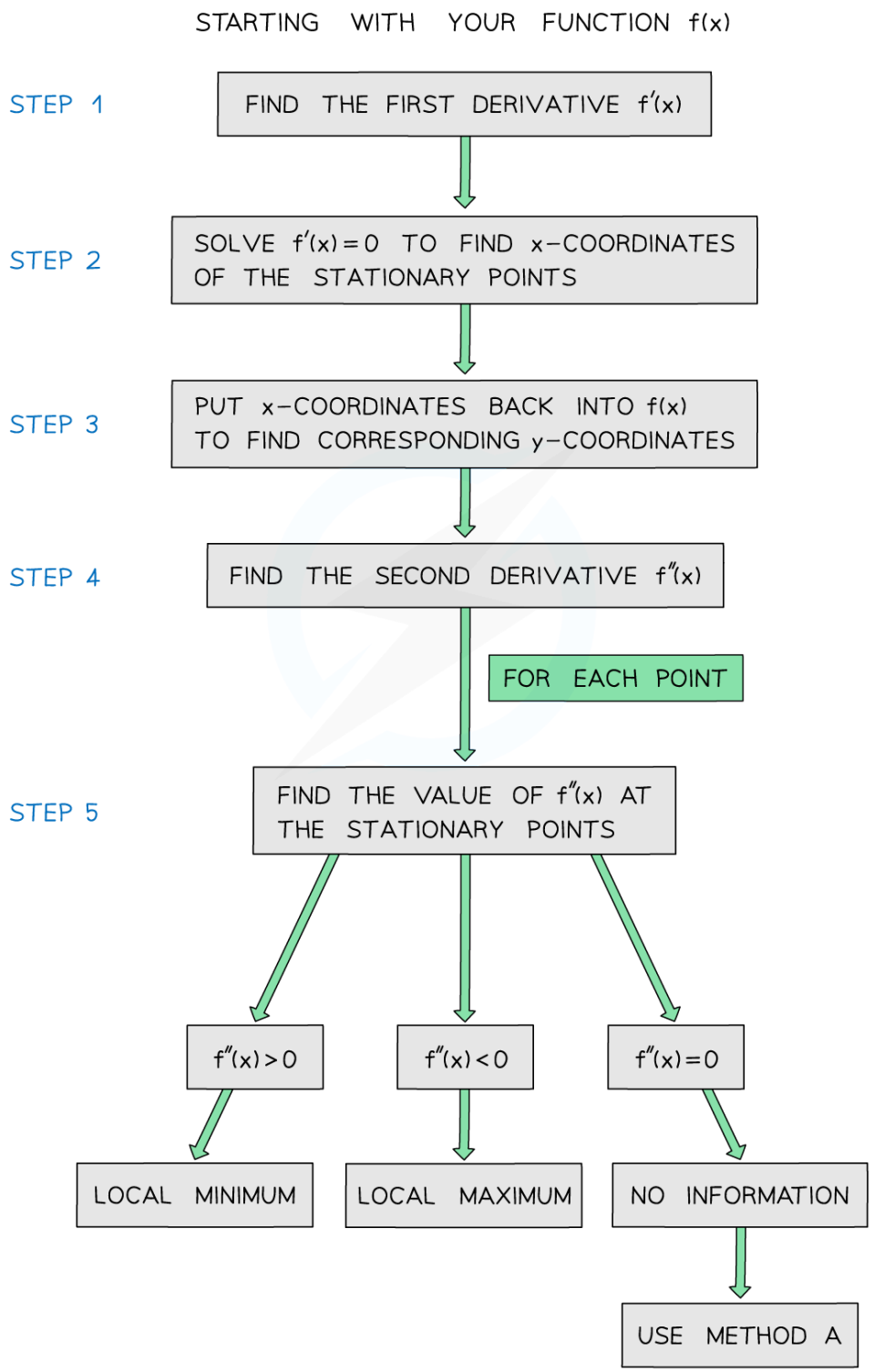
Your notes

METHOD B: DETERMINING NATURE OF STATIONARY POINTS USING SECOND DERIVATIVE



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Examiner Tips and Tricks

- Usually using the second derivative (Method B above) is a much quicker way of determining the nature of a stationary point.
- However, if the second derivative is zero it tells you nothing about the point.
    - In that case you will have to use Method A (which **always** works – see the Worked Example).

Worked Example

Your notes

Find the stationary points of $f(x) = x^3(3x^2 - 20)$, and determine the nature of each.

$f(x) = 3x^5 - 20x^3$ ← EXPAND THE BRACKETS

$f'(x) = 15x^4 - 60x^2$ ← STEP 1: FIND FIRST DERIVATIVE

$15x^4 - 60x^2 = 0$ ← STEP 2: SOLVE $f'(x) = 0$

$x^4 - 4x^2 = 0$

$x^2(x^2 - 4) = 0$

$x^2(x+2)(x-2) = 0$

$x = 0, -2, 2$ ← x-COORDINATES OF THE TURNING POINTS

IF $x = 0$, $y = (0)^3(3(0)^2 - 20) = 0$

IF $x = -2$, $y = (-2)^3(3(-2)^2 - 20) = 64$

IF $x = 2$, $y = (2)^3(3(2)^2 - 20) = -64$

← STEP 3: FIND CORRESPONDING y-COORDINATES

THE STATIONARY POINTS ARE
$(-2, 64)$, $(0, 0)$ AND $(2, -64)$

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START WITH METHOD B

$f''(x) = 60x^3 - 120x$

STEP 4: FIND SECOND DERIVATIVE

$f''(-2) = 60(-2)^3 - 120(-2) = -240 < 0$

$f''(0) = 60(0)^3 - 120(0) = 0$

$f''(2) = 60(2)^3 - 120(2) = 240 > 0$

STEP 5: TEST SECOND DERIVATIVE AT STATIONARY POINTS

(−2,64) IS A LOCAL MAXIMUM

(2,−64) IS A LOCAL MINIMUM

NOTE THAT AT (0,0) THE SECOND DERIVATIVE IS ZERO, AND SO DOESN'T TELL US WHAT THE POINT IS!

SWITCH TO METHOD A

TEST $f'(x)$ AT $x = -1$ AND $x = 1$

STEP 4: COMPARE FIRST DERIVATES TO LEFT AND RIGHT OF STATIONARY POINT

$f'(-1) = 15(-1)^4 - 60(-1)^2 = -45 < 0$

$f'(1) = 15(1)^4 - 60(1)^2 = -45 < 0$

STEP 5: INTERPRET RESULT

$f'(x)$ HAS SAME SIGN ON BOTH SIDES OF (0,0)

(0,0) IS A POINT OF INFLECTION

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Your notes

Your notes

Sketching gradient functions

How do I sketch the gradient function from the original function?

- Using your knowledge of gradients and derivatives you can use the graph of a function to sketch the corresponding gradient function
- The behaviour of a function tells you about the behaviour of its gradient function

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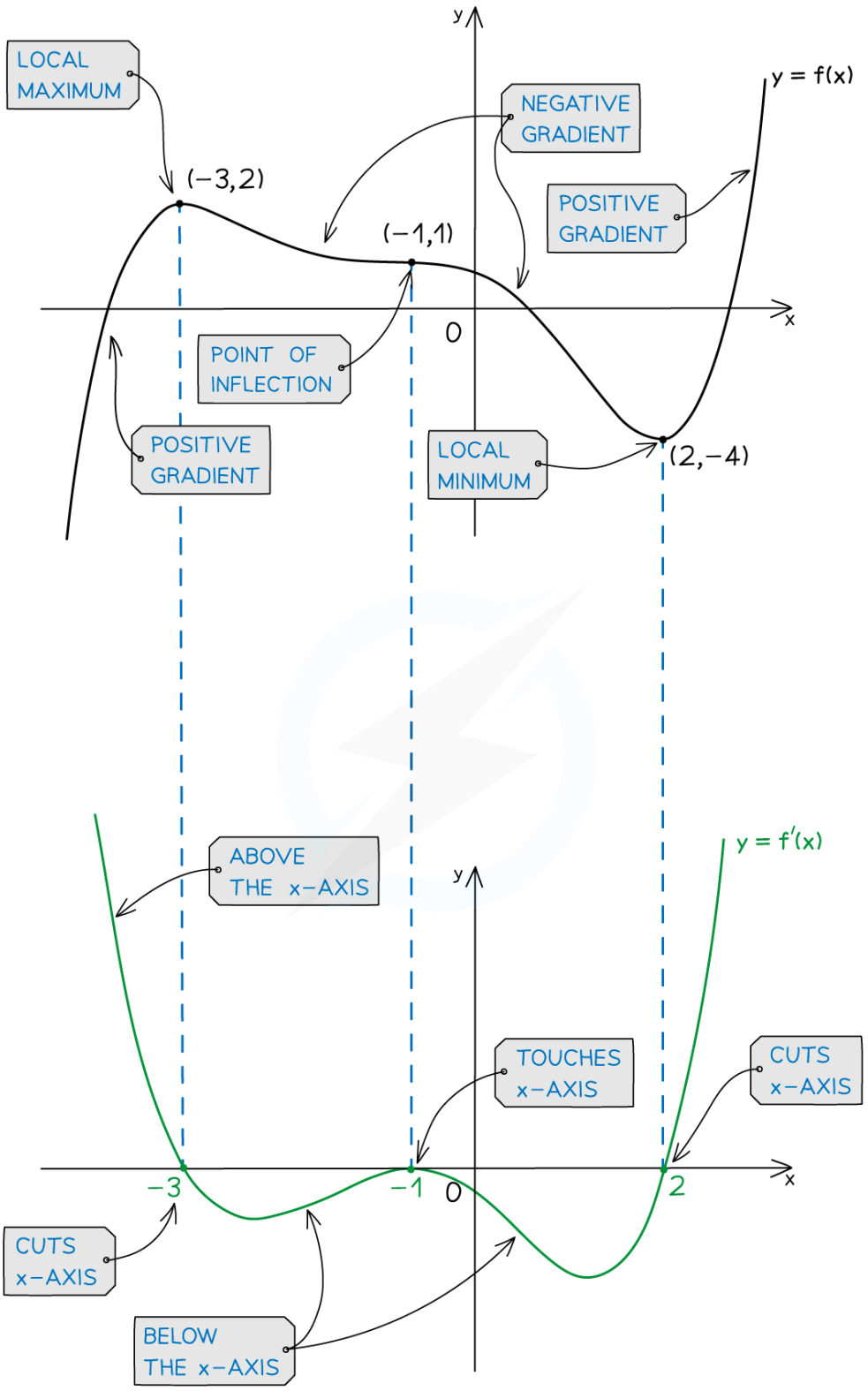

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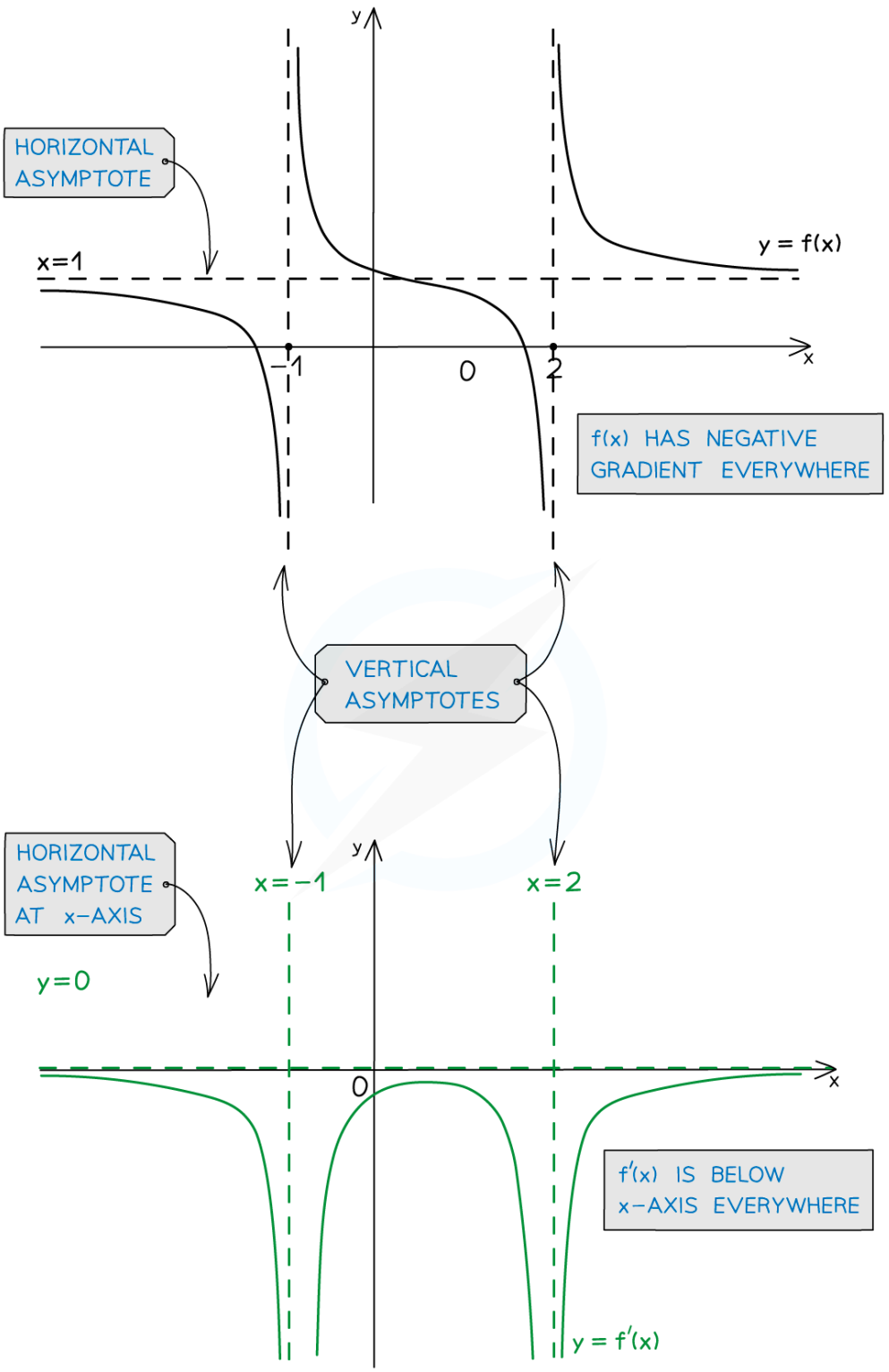

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Examiner Tips and Tricks

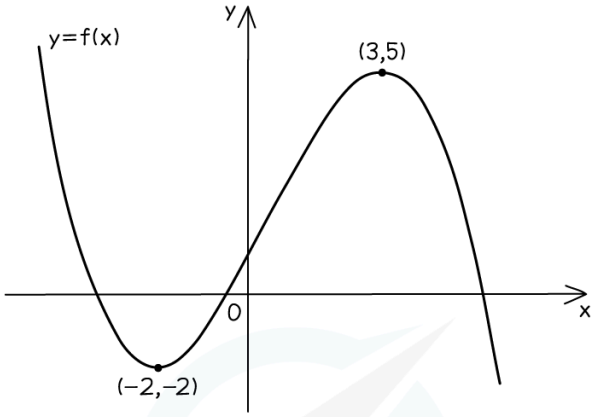
- If **f(*x*)** is a smooth curve then **f'(*x*)** will also be a smooth curve.
- Take what you know about **f'(*x*)** (based on the table above) and then 'fill in the blanks' in between.
- If all you have is the graph of **f(*x*)** you will not be able to specify the coordinates of the *y*-intercept or any stationary points of **f'(*x*).**
- Be careful – points where **f(*x*)** cuts the *x*-axis don't tell you anything about the graph of **f'(*x*)**!

Your notes

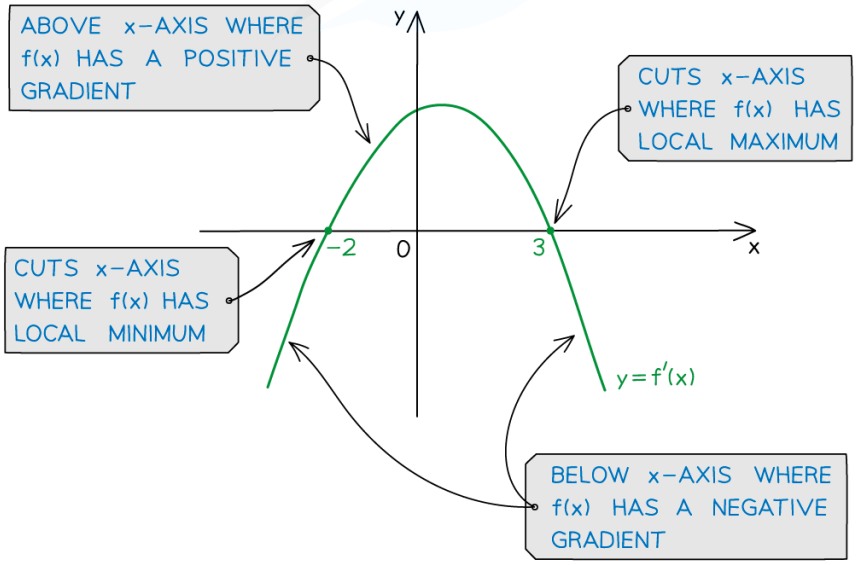
Worked Example

Your notes

The following diagram shows the graph of $y = f(x)$



Sketch the graph of $y = f'(x)$.



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Your notes

Modelling with differentiation (optimisation)

How do I find the maximum or minimum values of models?

- Derivatives can be calculated for any variables – not just ***y*** and ***x***
- In every case the derivative is a formula giving the rate of change of one variable with respect to the other variable

$$A = 4\pi r^2 \qquad \frac{dA}{dr} = 8\pi r$$

RATE OF CHANGE OF A WITH RESPECT TO r

$$T = 2\pi\sqrt{\frac{L}{g}} = \frac{2\pi}{\sqrt{g}}L^{\frac{1}{2}}$$

$$\frac{dT}{dL} = \frac{\pi}{\sqrt{g}}L^{-\frac{1}{2}} \qquad \frac{dT}{dL} = \frac{\pi}{\sqrt{gL}}$$

RATE OF CHANGE OF T WITH RESPECT TO L

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- Differentiation can be used to find maximum and minimum points of a function (see Stationary Points)
- Therefore it can be used to solve maximisation and minimisation problems in modelling questions

Your notes

e.g. AT TIME t SECONDS THE HEIGHT OF A BALL ABOVE THE GROUND IN METRES IS GIVEN BY FORMULA

$$h = 20t - 5t^2$$

USE DIFFERENTIATION TO FIND THE MAXIMUM HEIGHT REACHED BY THE BALL.

$$\frac{dh}{dt} = 20 - 10t$$

$$20 - 10t = 0$$

$$t = 2$$

THE GRAPH OF $20t - 5t^2$ IS A "DOWN" PARABOLA. THE MAXIMUM VALUE OCCURS WHEN $\frac{dh}{dt} = 0$

WHEN $t = 2$,

$$h = 20(2) - 5(2)^2 = 20 \text{ METRES}$$

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Examiner Tips and Tricks

Exam questions on this topic will often be divided into two parts:

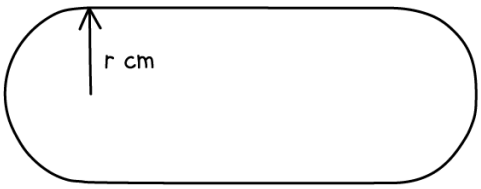
- First a 'Show that...' part where you derive a given formula from the information in the question
- And then a 'Find...' part where you use differentiation to answer a question about the formula

Even if you can't answer the first part you can still use the formula to answer the second part.

Worked Example

Your notes

A large allotment bed is being designed as a rectangle with a semicircle on each end, as shown in the diagram below.



The total area of the bed is to be 100π m^2.

a) Show that the perimeter of the bed is given by the formula $P = \pi\left(r + \frac{100}{r}\right)$

b) For what value of r is the rate of change of P with respect to r equal to $-\frac{9\pi}{16}$ metres per metre?

c) Find the perimeter of the bed for the value of r found in part (b).

a)

THE WIDTH OF THE RECTANGLE IS 2r AND THE LENGTH IS L. THE AREA OF THE BED IS

$$A = \frac{1}{2}\pi r^2 + 2rL + \frac{1}{2}\pi r^2 = \pi r^2 + 2rL$$

SEMICIRCLE

RECTANGLE

SEMICIRCLE

THE AREA IS 100π, SO

$$\pi r^2 + 2rL = 100\pi$$

$$2rL = 100\pi - \pi r^2$$

$$L = \frac{50\pi}{r} - \frac{\pi}{2}r$$

USE GIVEN AREA TO WRITE L IN TERMS OF r

Your notes

THE PERIMETER OF THE BED IS

$$P = \pi r + \pi r + 2L = 2\pi r + 2L$$

SEMICIRCLES

$$P = 2\pi r + 2\left(\frac{50\pi}{r} - \frac{\pi}{2}r\right)$$

$$= 2\pi r + \frac{100\pi}{r} - \pi r$$

$$= \pi r + \frac{100\pi}{r}$$

$$P = \pi\left(r + \frac{100}{r}\right)$$

SUBSTITUTE VALUE FOR L INTO PERIMETER FORMULA

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b) $P = \pi r + 100\pi r^{-1}$

$$\frac{dP}{dr} = \pi - 100\pi r^{-2} = \pi\left(1 - \frac{100}{r^2}\right)$$

DIFFERENTIATE P WITH RESPECT TO r

$$\pi\left(1 - \frac{100}{r^2}\right) = -\frac{9\pi}{16}$$

$$1 - \frac{100}{r^2} = -\frac{9}{16}$$

$$\frac{100}{r^2} = \frac{25}{16}$$

$$r^2 = 64 \implies r = \pm 8$$

BECAUSE r IS A LENGTH IT CANNOT BE NEGATIVE

$r = 8$ METRES

c) WHEN $r = 8$

$$P = \pi\left(8 + \frac{100}{8}\right)$$

$P = \frac{41\pi}{2}$ METRES $(\approx 64.4\text{ m})$

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Your notes

Connected Rates of Change

What are connected rates of change?

- In situations involving more than two variables you can use the **chain rule** to connect multiple rates of change into a single equation

e.g. THE VOLUME V ,OF, A CUBE IS INCREASING AT A CONSTANT RATE OF 3 cm^3 PER SECOND. ASSUMING THAT THE SHAPE REMAINS CUBICAL AT ALL TIMES, FIND THE RATE OF CHANGE OF THE SIDE LENGTH, S, IN cm PER SECOND AT THE MOMENT WHEN S = 5 cm.

$$\frac{dV}{dt} = 3 \text{ cm}^3/\text{sec}$$

WE KNOW THIS

WE WANT TO KNOW THIS

BY THE CHAIN RULE $\frac{dV}{dt} = \frac{dV}{ds} \times \frac{ds}{dt}$

USE THE CHAIN RULE TO CONNECT WHAT YOU KNOW AND WHAT YOU WANT TO KNOW

FOR A CUBE, $V = s^3$

SO $\frac{dV}{ds} = 3s^2$

USE DIFFERENTIATION TO GET THE 'MISSING' DERIVATIVE FROM THE CHAIN RULE EQUATION

SO $\frac{ds}{dt} = \frac{\frac{dV}{dt}}{\frac{dV}{ds}} = \frac{3}{3s^2} = \frac{1}{s^2}$

REARRANGE

WHEN s =5,

$$\frac{ds}{dt} = \frac{1}{5^2} = \frac{1}{25} \text{ cm/sec}$$

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Examiner Tips and Tricks

- These problems can involve a lot of letters – be sure to keep track of what they all refer to.

- Be especially sure that you are clear about which letters are **variables** and which are **constants** – these behave very differently when differentiation is involved!

Your notes

Worked Example

Your notes

In a manufacturing process, plastic spheres are produced in such a way that the volume, V, of a sphere increases at a constant rate of 10 cm³ per second. Find the rate of change of the surface area, A, of a sphere at the moment when the surface area is equal to 32π cm².

$\frac{dV}{dt} = 10 \text{ cm}^3/\text{sec}$

WE KNOW THIS

WE WANT TO KNOW THIS

BY THE CHAIN RULE $\frac{dV}{dt} = \frac{dV}{dA} \times \frac{dA}{dt}$

USE THE CHAIN RULE TO CONNECT WHAT YOU KNOW AND WHAT YOU WANT TO KNOW

$V = \frac{4}{3}\pi r^3$ (WHERE r IS THE RADIUS)

WRITE V IN TERMS OF A

$A = 4\pi r^2$

$A^{\frac{3}{2}} = (4\pi r^2)^{\frac{3}{2}} = 8\pi^{\frac{3}{2}} r^3 = 8\pi\sqrt{\pi}\, r^3$

$\frac{8\pi\sqrt{\pi}\, r^3}{6\sqrt{\pi}} = \frac{4}{3}\pi r^3$ SO $V = \frac{1}{6\sqrt{\pi}} A^{\frac{3}{2}}$

AND $\frac{dV}{dA} = \frac{3}{2} \times \frac{1}{6\sqrt{\pi}} A^{\frac{3}{2}-1} = \frac{1}{4\sqrt{\pi}} A^{\frac{1}{2}}$

DIFFERENTIATE TO FIND THE 'MISSING' DERIVATIVE

SO $\frac{dV}{dt} = \frac{dV}{dA} \times \frac{dA}{dt}$

SUBSTITUTE AND REARRANGE

$10 = \frac{1}{4\sqrt{\pi}} A^{\frac{1}{2}} \left(\frac{dA}{dt}\right)$

AND $\frac{dA}{dt} = \frac{40\sqrt{\pi}}{A^{\frac{1}{2}}} = 40\sqrt{\frac{\pi}{A}}$

WHEN $A = 32\pi$,

$\frac{dA}{dt} = 40\sqrt{\frac{\pi}{32\pi}} = \frac{40}{\sqrt{32}} = 5\sqrt{2} \text{ cm}^2/\text{sec}$

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Your notes